

NADA ALJAAFARI

Data & Business Analyst | Saudi Arabia

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PROFESSIONAL SUMMARY

Data and business analyst with an M.S. in Business Analytics (First Honors, KFUPM) and a B.S. in Management Information Systems. Builds end-to-end analyses in Python, SQL and R, from data cleaning and forecasting to classification, clustering and optimization, and turns the results into recommendations decision-makers can act on.

EDUCATION

Master of Science in Business Analytics, First Honors | KFUPM | 2024 – 2026

Bachelor of Science in Management Information Systems | King Faisal University | 2016 – 2021

EXPERIENCE

Executive Secretary | Al-Kifah Academy | Sep 2022 – Nov 2022

- Prepared reports and process documentation for senior leadership; coordinated daily operations and tracked requirements, flagging process inefficiencies.

Business Analyst Intern | Saudi Irrigation Organization | Mar 2021 – Jan 2022

- Evaluated internal systems through structured testing and requirements validation; presented redesign recommendations to senior managers.
- Surveyed employees on website usability and converted the results into documented requirements and visual reports.

ANALYTICS PROJECTS

Emergency Department Demand & Patient-Flow Analytics

Team of 4 | Python, SARIMA, Prophet, Random Forest, SMOTE, Gaussian Mixture Model

- Owned severity classification: on highly imbalanced data (20 Severity-1 training cases), Random Forest + SMOTE raised macro F1 from 0.32 to 0.48 over a Decision Tree, catching rare critical cases the higher-accuracy model missed.
- Team results on 90,036 ER visits (Jubail hospital, 2022): SARIMA forecast daily demand (MAPE 21.3%, beating Prophet), a 10 AM – 4 PM arrival peak, and three patient-flow clusters (GMM).

Saudi Arabia as a Medical Tourism Hub (Master's Capstone)

KFUPM | Python, statistical testing, classification, SMOTE

- Compared 79,000 patient records for Saudi and non-Saudi patients using t-tests, correlation analysis and classification models.
- Found near-identical costs (SAR 49.8K vs. 50.2K); even with SMOTE balancing, classifiers could not separate the two groups by clinical profile (59.7% vs. 50% chance), evidence of consistent care.
- Translated the findings into five recommendations aligned with Vision 2030, including simpler medical-visa pathways and multilingual patient services.

TikTok 30-Day Views Prediction (Kaggle Competition)

Python | Random Forest, XGBoost, AdaBoost, feature engineering

- Engineered 19 engagement and creator features from first-five-day data and trained a weighted ensemble of three models (validation $R^2 \approx 0.93$); placed in the top 27 on the competition leaderboard.

Mostakan Tea Room: Marketing Budget Optimization

Team of 4, real client | Linear programming (Excel Solver, Python), sensitivity analysis

- Analyzed 2023–2025 sales and order data for an Al-Ahsa café to confirm a decline in demand, then built a linear-programming model allocating the marketing budget across five social channels; sensitivity analysis supported a moderate budget increase with diversified spend.

Spam Detection Web Application

Python | NLP, TF-IDF, SMOTE, scikit-learn, Flask

- Built a text classifier on ~5,600 imbalanced messages; Naive Bayes caught 131 of 150 spam with only 10 false positives and was deployed in a Flask app.

TECHNICAL SKILLS

Programming: Python (pandas, scikit-learn, XGBoost, Prophet), SQL, R, HTML/CSS, Flask

Analytics: Regression, classification, clustering, time-series forecasting, linear programming, hypothesis testing, NLP

Tools: Microsoft Excel, PowerPoint, Visio, Jupyter, Git/GitHub

Business Analysis & UX: Requirements gathering, systems analysis, process documentation, usability testing, wireframing

Certifications: Google UX Design Certificate (2024) | UX Design (Accelerated), Misk Skills (2023) | PMP Training (2021) | Commercial Marketing Management Training (2021)

Languages: Arabic (native), English (professional; IELTS Academic 7.0, CEFR C1)